Mechanisms of star fruit-induced acute renal failure.

We have previously discovered that star fruit can induce oliguric acute renal failure. To investigate the mechanisms of star fruit-associated acute oxalate nephropathy, the nephrotoxic effect of star fruit was examined in both cellular experiments and animal models. We evaluated renal function, pathological changes in kidney tissues and apoptotic effects using terminal deoxynucleotidyl transferase nick-end labeling (TUNEL) assay in four groups of rats -- a control group (CG), fed with tap water (1); a star fruit group (SG), fed with star fruit juice naturally containing 0.2M oxalate (2); and oxalate groups (OxG), fed with 0.2M (3) or 0.4M (4) oxalate solution. The effects of both star fruit juice and oxalate on MDCK cells were also analyzed by flow cytometry. We found that the mean creatinine clearance was significantly lower in the SG, 0.2M OxG and 0.4M OxG. Dose-dependent apoptotic effects were evident from the TUNEL assay, and flow cytometry analysis of treated MDCK cells showed dose- and time-dependent effects. Our findings suggest that star fruit juice produces acute renal injury, not only through the obstructive effect of calcium oxalate crystals, but also by inducing apoptosis of renal epithelial cells, which may be caused by the levels of oxalate in the fruit.